MILITARY SPECIFICATION

TRANSMITTING SET

TELEMETRIC DATA

AN/DKT-58(V1)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the AN/DKT-58(V1) Telemetric Data Transmitting Set, referred to herein as the TMX section.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-794	Parts and Equipment Procedures, for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1695	Environments, Working, Standards for.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

Code of Federal Regulations (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Inter-range Instrumentation Group (IRIG)

IRIG Document 106	Telemetry Standards.

(Application for copies should be addressed to the Secretariat, Range Commanders Council, White Sands Missile Range, New Mexico 88002.)

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DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

2604830	Housing, Telemetric Section Type I.
639AS4900	Transmitting Set Telemetric, Data, AN/DKT-58(V1).
639AS4901	Transmitting Set Subassembly.
639AS4903	Transmitting Set Interconnection Diagram (AN/DKT-58(V1)).
639AS4915	Circuit Card Assembly Signal Conditioner.
639AS6167	Thermal Battery/Transmitter Cable Assembly.
639AS6203	Modulation Commutator Assembly (28 V).
639AS6204	Modulation Commutator Assembly (15 V).
SA 494813	Fixture, Vibration SW, TM (DR 2604830).

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-1322	Reliability Program Provisions for Production of Sidewinder AIM-9.
AS-1986	Antenna, Stripline.
AS-1989	Thermal Battery.
AS-1990	Transmitter, Telemetry.
AS-1991	Power Supply, Airborne.
AS-4462	Commutator, High Speed, Airborne.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

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Naval Weapons Center
(Code Ident 12934)

352-213-83 — Electromagnetic Environmental Effects (E^3) Systems Specification for the AN/DKT-58 Telemetric Section.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The TMX section is used to sample and transmit flight performance information for a 5-inch guided missile.

3.1.1 Major components. The TMX section consists of the following major components:

a. Housing, Telemetric Section Type I (see Drawing 2604830).

b. Transmitting Set, Telemetric Data, AN/DKT-58(V1) (see Drawing 639AS4900).

c. Transmitting Set Subassembly (see Drawing 639AS4901).

d. Circuit Card Assembly Signal Conditioner (see Drawing 639AS4915).

e. Antenna (see AS-1986).

f. Commutator (see AS-4462).

g. Commutator Assemblies (see Drawing 639AS6203 or Drawing 639AS204).

h. Thermal Battery Assembly (see AS-1989).

i. Thermal Battery/Transmitter Cable Assembly (see Drawing 639AS6167.

j. Transmitter (see AS-1990).

k. Power Supply, Airborne (see AS-1991).

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3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 TMX section subassembly. The major components, parts, and necessary wiring shall be assembled and connected into a TMX section subassembly in accordance with Drawing 639AS4901 and associated documents. The subassembly performance requirements shall be in accordance with this specification.

3.2.1.2 Source current transmitter off. The source current with the transmitter off shall be not greater than 350 milliamperes (mA).

3.2.1.3 Source current transmitter on. The source current with the transmitter on shall be not greater than 1.5 amperes (A).

3.2.1.4 Transmitter power output. The transmitter power output shall be in accordance with the requirements of AS-1990.

3.2.1.5 Transmitter center frequency. Unless otherwise specified in the contract or purchase order (see 6.2.1), the transmitter center frequency tolerance shall be ±0.002 percent of the specified frequency, and stability about the actual center frequency shall be ±0.003 percent with or without modulation.

3.2.1.6 Data and synchronization outputs, deviation. Channel amplitudes as specified herein shall be observed during the acquisition window period (see 6.3.1). The noise on all channels shall be less than 1 percent full scale. The subassembly shall accept 43 data inputs and transmit them sequentially, followed by five channels of synchronization and calibration voltages in accordance with IRIG Document 106 and the applicable pulse amplitude modulation (PAM) output format as shown in Figure 1. Channel 44 shall be zero reference and shall correspond to a transmitter deviation between 240 and 270 kilohertz (kHz). Channel 46 shall be 100 percent reference and shall correspond to a transmitter deviation in the direction opposite the zero reference. Channel 48 shall be 49 to 51 percent on the scale as specified herein. Channel input impedances, offsets, and other attributes shall be in accordance with AS-4462. Total peak-to-peak transmitter deviation with any normal input signal shall be 480 to 540 kHz.

3.2.1.7 Risetime and switching transient overshoot. Risetime on the leading edge of any channel shall be greater than 2.0 microseconds (µsec). Switching transient overshoot within any synchronization or data channel pulses shall be not greater than 5 percent of the scale (see 3.2.1.6), and the duration shall be not greater than 1 µsec (see Figure 2 and 6.3).

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3.2.1.8 Squib tests. With the thermal relay in the unfired position, the amplitude of channel 40 shall be -1.0 to -1.5 volts direct current (Vdc). Thermal battery squib (match) resistance shall be between 0.55 and 0.70 ohms when measured between pins 43 and 49 of plug P1. Launch latch resistance shall be measured between pins 44 and 49 of P1 to verify diode direction. Thermal battery and launch latch resistance shall be measured with a squib meter current limited to 100 mA or less.

3.2.1.9 Airborne power supply. The airborne power supply shall provide only ±15 Vdc.

3.2.2 TMX section. The basic TMX subassembly shall be interconnected and installed into the housing as specified herein. The antenna shall be installed on the housing and connected to the transmitter output cable. Upon the application of 22 to 34 Vdc to plug P1-42, the TMX section shall transmit telemetric data as specified herein.

3.2.2.1 Commutator outputs (transmitted). The application of a voltage to a specific pin on plug P1 as specified herein shall cause the transmitted PAM channel outputs of the TMX section to vary.

3.2.3 Reliability. When specified in the contract or purchase order (see 6.2.2) the contractor shall establish and implement a reliability program plan in accordance with AD-1322.

3.2.3.1 Mean-time-between-failure (MTBF) demonstration. Unless otherwise specified in the contract or purchase order (see 6.2.1), telemetry (TM) supplied in accordance with this specification shall demonstrate a specified MTBF (θ_o) of 50 hours when tested in accordance with MIL-STD-781, Test Plan IVC.

3.2.3.1.1 MTBF performance requirements. The TM shall be operated during the heating subcycle. Operation shall consist of the application of normal external power. The thermal battery shall not be activated. Performance shall be monitored in a manner to detect any cessation of function. Parameter limits need not be verified during this operation.

3.2.3.1.2 MTBF determination. The TM operating time shall be used in MTBF determination. Each TM operating time shall be not less than one-half of the average operating time of all TMs which were utilized in the MTBF demonstration. TM operating time shall be that time during the thermal cycle when the TM is energized.

3.2.4 Environmental. The TMX section shall show no evidence of physical damage and shall meet all requirements as specified herein, during and after exposure to the following environmental conditions:

a. Temperature shock (operating).

b. High temperature (operating).

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c. Captive flight vibration (operating).

d. Transportation shock (nonoperating).

e. Tip-off shock (operating).

f. Temperature/humidity/altitude (operating).

g. Acoustics noise (operating).

h. Transportation vibration (nonoperating).

i. Salt fog (nonoperating).

j. Aerodynamic heat (nonoperating).

k. Degree of enclosure (nonoperating).

3.2.5 Burn-in and screening. Each TMX section shall be subjected to a minimum of three cycles of burn-in and screening as shown in Figure 4.

3.2.6 Vibration. Vibration shall be as follows.

3.2.6.1 Sinusoidal. Levels of sinusoidal vibration up to and including 5 gravity units (g), or 0.06 inch double amplitude excursion, whichever is less, over a frequency spectrum from 5 to 2,000 hertz, in all axes.

3.2.6.2 Random. Random vibration with a power spectral density as specified in Figure 3.

3.3 Design and construction.

3.3.1 Production drawings. The TMX section major components and parts shall be manufactured, tested, inspected, and assembled in accordance with the requirements, processes, and procedures of Drawing 639AS4900 and associated documents.

3.3.1.1 Serial number. Serial numbers shall be as specified in the contract or purchase order (see 6.2.1).

3.3.2 Standards of manufacture.

3.3.2.1 Working environment. Unless otherwise specified in the contract or purchase order (see 6.2.1), the facilities used in the manufacturing, testing, and inspection of the TMX section shall meet the minimum levels and requirements of MIL-STD-1695.

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3.3.2.2 Production processing. The TMX section shall be manufactured, inspected, and tested under conditions and procedures as specified herein.

3.3.2.3 Electromagnetic environment. The contractor shall establish controls that preclude any degradation of TMX section integrity to electromagnetic environmental effects during the manufacturing process. The TMX section shall meet the electromagnetic environment requirements of 352-213-83.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.5 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. Soldering processes shall be in accordance with WS-6536.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

c. MTBF demonstration inspections (see 4.5.4).

4.2 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.3), a first article sample consisting of four TMX sections shall be delivered to the testing activity designated in the contract or purchase order for first article testing. The first article sample shall be manufactured using the same methods, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample shall be at the risk of the contractor. After approval of the first article sample, changes in materials, parts, processes, procedures, or design shall require prior written approval from the contracting activity.

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4.2.1 First article inspection. First article inspections shall be subjected to the tests and inspections of Table I, in the sequence shown. The first article sample shall be in accordance with 4.2.

Table I. First article inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (all TMX subassemblies)		
Visual examinations	3.3.1, 3.5	4.5.6
Data and synchronization channel outputs	3.2.1.6	4.5.1.7
Deviation	3.2.1.6	4.5.1.8
Risetime and switching, transient overshoot	3.2.1.7	4.5.1.9
Activation resistance	3.2.1.8	4.5.1.10
Power supply	3.2.1.9	4.5.1.11
Group II (all TMX sections)		
Performance tests	...	4.2.3
Temperature shock	3.2.4a.	4.5.5.1
Post tests	...	4.2.3
Group III (Unit 1)		
High temperature	3.2.4b.	4.5.5.2
Post tests	...	4.2.3
Captive flight vibration	3.2.4c.	4.5.5.3
Post tests	...	4.2.3
Transportation shock	3.2.4d.	4.5.5.4
Post tests	...	4.2.3
Tip-off shock	3.2.4e.	4.5.5.5
Post tests	...	4.2.3
Temperature/humidity/altitude	3.2.4f.	4.5.5.6
Post tests	...	4.2.3
Acoustics	3.2.4g.	4.5.5.7
Performance tests	...	4.2.3
Group IV (Unit 2)		
Temperature/humidity/altitude	3.2.4f.	4.5.5.6
Post tests	...	4.2.3
Captive flight vibration	3.2.4c.	4.5.5.3
Post tests	...	4.2.3
Transportation vibration	3.2.4h.	4.5.5.8
Post tests	...	4.2.3
Salt fog	3.2.4i.	4.5.5.9
Performance test	...	4.2.3

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Table I. First article inspections (continued).

Examination or test	Requirement paragraph	Method paragraph
Group V (Unit 3)		
High temperature	3.2.4b.	4.5.5.2
Post tests	...	4.2.3
Transportation shock	3.2.4d.	4.5.5.4
Post tests	...	4.2.3
Degree of enclosure	3.2.4k.	4.5.5.11
Performance tests	...	4.2.3
MTBF demonstration	3.2.3.1.1	4.5.4
Group VI (Unit 4)		
Captive flight vibration	3.2.4c.	4.5.5.3
Post tests	...	4.2.3
Tip-off shock	3.2.4e.	4.5.5.5
Post tests	...	4.2.3
Aerodynamic heat	3.2.4j.	4.5.5.10
Performance tests	...	4.2.3
MTBF demonstration	3.2.3.1.1	4.5.4

4.2.1.1 First article sample designation. The four first article sample TMX sections shall be consecutively numbered 1 through 4. Each numbered unit shall be subjected to the tests corresponding to the unit number as specified in Table I.

4.2.2 First article acceptance criteria. Failure of any TMX section to pass any of the first article tests as specified herein shall cause rejection of the first article sample.

4.2.3 Performance tests/post tests. The performance tests/post tests shall be performed in accordance with the test methods of 4.5 and shall consist of verification of the following:

a. Determine TMX section turns on and off properly (see 4.5.1.2).

b. Source current transmitter off (see 4.5.1.3).

c. Source current transmitter on (see 4.5.1.4).

d. Transmitter power output (see 4.5.1.5).

e. Transmitter center frequency (see 4.5.1.6).

f. Channel voltage levels (see 4.5.1.7b).

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g. Verify channels 2, 10, 18, 26, 34, and 42 for conformance (see 4.5.1.7.c).

h. Observe PAM on scope (see 4.5.1.7d).

i. Transmitter upper band edge frequency, positive peak (see 4.5.1.8c).

j. Transmitter lower band edge frequency, negative peak (see 4.5.1.8d).

k. Peak-to-peak deviation (see 4.5.1.8e).

l. Rise time and switching transient overshoot (see 4.5.1.9).

m. Squib test (see 4.5.1.10).

n. Airborne power supply output (see 4.5.1.11).

4.3 Quality conformance inspections. All TM section lots offered for acceptance testing shall be subjected to and shall pass the quality conformance inspections specified in Table II.

Table II. Quality conformance inspections.

Examination or test	Requirement paragraph	Method paragraph
Group A (all TMX sections)		
Table I, Group I tests	Table I, Group I	Table I, Group I
Burn-in and temperature cycling	3.2.5	4.5.3
Vibration screening	3.2.6	4.5.3.2
Performance tests	---	4.2.3
Group B (sample sections)		
Temperature shock	3.2.4a	4.5.5.1
MTBF	3.2.3.1.1	4.5.4
Performance tests	---	4.2.3

4.4 Measuring and test equipment.

4.4.1 Calibration and maintenance. Unless otherwise specified in the contract or purchase order (see 6.2.1), all measuring and test equipment used by the contractor for performance of the inspections specified

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herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order (see 6.2.2), certification to this effect shall accompany reports of first article and quality conformance tests and inspections.

4.4.2 Test equipment. Unless otherwise specified in the contract or purchase order (see 6.2.1), test equipment shall be approved by the contracting activity prior to Government acceptance of produced items. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements under the conditions and to the accuracies specified herein.

4.4.2.1 Test equipment accuracy. All test equipment used in the performance of the specified electrical tests shall have an accuracy of at least one-fifth the tolerance for the variable to be measured.

4.4.2.2 Test equipment required. The following test equipment shall be used in the performance of the tests specified herein, in accordance with 4.4.2.

a.	Antenna test facility reflection level:	-30 decibels (dB) minimum within an 18-inch radius quiet zone.
	(1) Angular positioning system accuracy:	±1.0 degree maximum error.
	(2) Data acquisition system accuracy:	±0.5 dB maximum accumulative error.
b.	Digital voltmeter:	0.1 percent maximum error.
c.	Frequency counter:	Counting rate to 125 kHz.
d.	Frequency counter:	Counting rate to 2300 megahertz (MHz).
e.	Network analyzer and polar plotter:	Hewlett-Packard Model 8410 (or equal).
f.	Oscilloscope:	2 MHz minimum vertical deflection bandwidth and with a delayed sweep time base, variable from 1 microsecond per division (µsec/div) to 20 µsec/div (minimum).
g.	Power supply:	28 Vdc, 2 A minimum.

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h.	Receiver:	Ultra high frequency (UHF) telemetry type, 1 MHz intermediate frequency, and direct current (dc) to 1 MHz (minimum) output response.
i.	Square wave generator:	5 volts positive swing minimum, and maximum rise time of 0.5 µsec at 50 kHz.
j.	Squib meter:	Current limited to 100 mA or less.
k.	Radio frequency (RF) wattmeter:	3 watts (W) full scale.
l.	RF attenuator:	40 dB, 5 W.
m.	PAM synchronizer:	EMR-515 or equal.

4.5 Test methods.

4.5.1 Electrical.

4.5.1.1 Test setup. Prior to TMX section checkout, the test setup shall be prepared as follows:

a. Install the programmable connector (see Figure 5) into P1 connector (see Drawing 639AS4903).

b. Connect the 40 dB attenuator between the transmitter output cable and the UHF receiver input. Alternatively, with the antenna installed and connected, place an RF absorbing collar around the antenna and connect the collar output to the receiver. Place the TMX section horizontally in a cradle.

WARNING

Do not attempt to radiate from the telemetry system through the air into a receiving antenna, as exposure to the generated RF field may cause physical discomfort and possible eye damage.

c. Major components which comprise the TMX section shall be listed on the major components list as shown in Figure 6 (see 3.1.1).

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4.5.1.2 On/off test. The TMX section shall turn on and off properly. To determine this, proceed as follows:

a. Apply +28 Vdc to pin 41 of P1; the TMX section shall turn on. Upon removal of +28 Vdc, the TMX section shall turn off.

b. With a screwdriver, turn the manual on/off switch on and off. This switch shall also turn the TMX section on and off.

c. At the conclusion of steps a. and b., record the results on the logbook quality assurance (QA) sheets as shown in Figure 6. Place the manual switch in the off position and turn on the TMX section with the +28 Vdc to pin 41.

4.5.1.3 Source current transmitter off. The source current shall be measured with the transmitter off. Results shall be recorded on the logbook QA sheets as shown in Figure 6. The source current shall meet the requirements of 3.2.1.2.

4.5.1.4 Source current transmitter on. The source current shall be measured with the transmitter on. Results shall be recorded on the logbook QA sheets as shown in Figure 6. The source current shall meet the requirements of 3.2.1.3.

4.5.1.5 Transmitter power output. The transmitter power output shall be measured and the results shall be recorded on the logbook QA sheets as shown in Figure 6. The power output shall meet the requirements of 3.2.1.4.

4.5.1.6 Transmitter center frequency. Transmitter center frequency shall be measured with 30 Vdc being supplied to the TMX section through the programmable connector and the PAM output pin P1-48 grounded. The results shall be recorded on the logbook QA sheets as shown in Figure 6. The TMX section shall meet the requirements of 3.2.1.5.

4.5.1.7 Data and synchronization channel outputs (TMX subassembly). Data channels shall be operable and in the correct order. To determine this, proceed as follows:

a. Supply 30 Vdc through the programmable connector shown in Figure 5.

b. Data channels shall be monitored via an EMR-515 (or equal) decommutator and each channel shall be monitored with a digital voltmeter (DVM) and oscilloscope. The voltage levels for each

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channel shall be recorded on the logbook QA sheets as shown in Figure 6. The voltage levels for each channel shall meet the requirements in accordance with Figure 6. The input to the DVM and oscilloscope shall be normalized so that full-scale shall be +2.500 ±0.050 Vdc and zero shall be -2.500 ±0.050 Vdc.

c. Channels 2, 10, 18, 26, 34 and 42 with variable discrete levels shall be verified for conformance with an oscilloscope. Verification shall be recorded by a check mark on the logbook QA sheets as shown in Figure 6.

d. Adjust the oscilloscope sweep rate for viewing one complete frame. Record the results on the logbook QA sheets as shown in Figure 6. The TMX shall meet the requirements of 3.2.1.6.

e. Verification tests shall be performed as follows:

(1) Place switch (S_1) (see Figure 5), in the normally open position. The channels listed in step c. shall measure 0.000 to -0.050 Vdc, equal to 0 percent PAM output.

(2) Place switch (S_1) (see Figure 5) in the normally closed position and switch (S_3) in the normally open position. Toggle switch (S_2) to state 1 and the channels listed in step c. shall measure 1.43 to 1.59 Vdc, equal to 81 percent PAM output. Toggle switch (S_2) to state 2 and the channels listed in step c. shall measure 0.43 to 0.46 Vdc, equal to 60 percent PAM output. Toggle switch (S_2) to state 3 and the channels listed in step c. shall measure -0.43 to -0.47 Vdc, equal to 41 percent PAM output. Toggle switch (S_2) to state 4 and the channels listed in step c. shall measure -1.39 to -1.53 Vdc, equal to 20 percent PAM output.

(3) Place switch (S_1) in the normally closed position and switch (S_3) in the normally closed position. Toggle switch (S_2) to state 1 and the channels listed in step c. shall measure 1.43 to 1.59 Vdc, equal to 82 percent PAM output. Toggle switch (S_2) to state 2 and the channels listed in step c. shall measure -0.43 to -0.47 Vdc, equal to 41 percent PAM output.

4.5.1.8 Deviation. Deviation shall be measured as follows:

a. Connect a high-speed decade counter in events per unit time mode to the UHF receiver output.

b. Turn the transmitter on with +28 Vdc to pin 41 of P1 and ground pins P1-49 and P1-50.

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c. The transmitter deviation shall be adjusted by selecting the proper ratio between the select resistors R93 and R94 shown on Drawing 639AS4915. Nominal range shall be 0 to 1 kilohms (Kohms) for R93 and 500 to 3 Kohms for R94.

d. Apply +2.5 Vdc to PAM pin 48 of P1 for a period not greater than 60 seconds, and record the positive peak deviation frequency of the upper band edge on the logbook QA sheets as shown in Figure 6.

e. Apply -2.5 Vdc to PAM pin 48 of P1 for a period not greater than 60 seconds, and record the negative peak frequency deviation of the lower band edge on the logbook QA sheets as shown in Figure 6.

f. Add the results of steps c. and d. together and record the peak-to-peak deviation on the logbook QA sheets as shown in Figure 6.

The deviation shall meet the requirements of 3.2.1.6.

4.5.1.9 Risetime and switching transient overshoot. Risetime switching transient overshoot shall be measured as follows:

a. Connect the oscilloscope to the UHF receiver output with a 1 MHz (minimum) intermediate frequency (IF) and video bandwidth, and disconnect the external synchronization. Turn the transmitter on with +28 through the programmable connector (see Figure 5) to pin 41 of P1.

b. Adjust the oscilloscope sweep rate to 1 µsec/div. Observe risetime and switching transient overshoot and record on the logbook QA sheets as shown in Figure 6.

The TMX section shall meet the requirements of 3.2.1.7.

4.5.1.10 Squib test. The squib test shall be performed as follows:

a. Observe channel 40 and verify that the thermal relay has not been energized. Record the results on the logbook QA sheets as shown in Figure 6.

b. Turn off primary power and remove the programmable connector.

c. Measure the thermal battery squib resistance between P1-43 and P1-49. Record the results on the logbook QA sheets as shown in Figure 6.

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d. Using a squib meter, measure the resistance between P1-44 and P1-49. Record the results on the logbook QA sheets as shown in Figure 6.

e. Reverse the squib meter leads and measure the resistance between P1-44 and P1-49. Record the results on the logbook QA sheets as shown in Figure 6.

f. Squib tests shall be performed prior to and after environmental, sample, and MTBF tests.

The subassembly shall meet the requirements of 3.2.1.8.

4.5.1.11 Airborne power supply outputs. The airborne power supply output shall be measured as follows:

a. Measure the +15 Vdc output by connecting a DVM between P1-45 and P1-49 through the programmable connector as shown in Figure 5. Record the results on the logbook QA sheets as shown in Figure 6.

b. Measure the -15 Vdc output by connecting a DVM between P1-46 and P1-49 through the programmable connector as shown in Figure 5. Record the results on the logbook QA sheets as shown in Figure 6.

The subassembly shall meet the requirements of 3.2.1.9.

4.5.2 TMX section. TMX section performance tests shall consist of the tests as specified in 4.5.1. To be acceptable, the TMX section shall meet the requirements of 3.2.1.

4.5.3 Burn-in and temperature cycling. Each TMX section shall be subjected to a minimum of three burn-in and temperature cycles in accordance with Figure 4.

4.5.3.1 Operation of TMX section. The TMX section shall be operated 30 minutes prior to the start of each heating cycle and shall continue to operate for 1.5 hours at +70 ±5°C in accordance with Figure 4. Operation shall consist of installing the programmable connector (see Figure 5) into P1 connector (see Drawing 639AS4903) and applying nominal external power. The thermal battery shall not be activated. Performance shall be monitored and the parameter limits shall meet the requirements of 4.2.3 at the commencement of each cooling subcycle.

4.5.3.2 Vibration screening. Each TMX section shall be subjected to the vibration screening profiles of Figure 7. Vibration shall be performed in each of the three mutually perpendicular axes at ambient temperature. The TMX section shall be operated during vibration by applying normal external power. Monitoring shall be conducted in a manner

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to detect any cessation of function. At the conclusion of vibration screening a complete performance test in accordance with 4.2.3 shall be performed.

4.5.3.3 Failure and corrective action. A TMX section failure that occurs during burn-in and temperature cycling shall be reworked and resubmitted for testing, beginning with the cycle preceding the cycle that the TMX failed. A failure during cycle 1 shall be resubmitted at the beginning of cycle 1. A TMX section failure that is detected during the performance test conducted after vibration screening shall be reworked and resubmitted to the test sequence at the beginning of test cycle 2 in accordance with 4.5.3.

4.5.4 MTBF demonstration. Following the tests of Group A, Table II, all TMX sections shall be assembled into an inspection lot in accordance with MIL-STD-105.

4.5.4.1 MTBF Demonstration sample selection. A random sample selection for MTBF demonstration testing of Group B, Table II will be conducted by a Government representative in accordance with the following schedule:

Number of units per lot	Sampling size
1-150	4
151-300	6
301-500	8

4.5.4.2 Lot storage. After lot test samples have been selected in accordance with 4.5.4.1, the remainder of the lot shall be stored in bonded storage pending acceptance or rejection of 4.5.4.4.

4.5.4.3 MTBF sample demonstration test performance.

4.5.4.3.1 Shock. The mechanical shock test shall be performed in accordance with 4.5.5.4.

4.5.4.3.2 MTBF demonstration testing. At the completion of 4.5.4.3.1, MTBF demonstration testing shall be conducted in accordance with Figure 8 to meet the test time requirements of Figure 9.

4.5.4.4 Lot acceptance criteria. A production lot shall be acceptable when the MTBF demonstration test samples of 4.5.4.1 meet the acceptance criteria of Figure 9 prior to reaching the reject criteria of Figure 9.

4.5.4.5 Refurbishment. Each production TMX section subjected to MTBF demonstration tests shall be refurbished when necessary and subjected to the performance tests as specified herein.

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4.5.5 Environmental.

4.5.5.1 Temperature shock (operating). The TMX section shall be tested in accordance with MIL-STD-810, Method 503.1, Procedure I, except that the Step 2 temperature shall be -40 degrees Celsius (°C). The tests of 4.2.3 shall be performed approximately 2 hours after the start of each temperature condition for all three cycles. Step 9 shall consist of the tests of 4.2.3.

4.5.5.2 High temperature (operating). The TMX section shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure I. Operating temperature shall be +60°C. Step 5 shall consist of the tests of 4.2.3 during the last 2 hours. Step 7 shall consist of the tests 4.2.3.

4.5.5.3 Captive flight vibration (operating). The TMX section shall be mounted on the test fixture (see Drawing SA 494813), and tested in accordance with MIL-STD-810, Method 514.2, Procedure IIA, except that the power spectral density shall be as specified in Figure 3 and the test time shall be 0.5 hour per axis. The free flight functional test shall not be performed. External power shall be applied during all phases of the test. The tests of 4.2.3 shall be performed during the last 15 minutes of each axis at each temperature. Post tests shall be performed in accordance with 4.2.3. Tests shall be conducted at -40°C and +60°C only.

4.5.5.4 Transportation shock (nonoperating). The TMX section shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-2 except a 50-g amplitude and a 21-millisecond (ms) time duration shall be used. Post tests shall be performed in accordance with 4.2.3.

4.5.5.5 Tip-off shock (operating). The TMX section shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-2 except a 50-g amplitude and a 4-ms time duration shall be used. Operation shall consist of the application of external power. Post tests shall be performed in accordance with 4.2.3.

4.5.5.6 Temperature-humidity-altitude (nonoperating). The TMX section shall be tested in accordance with MIL-STD-810, Method 518.1, Procedure I, as modified in accordance with Figure 10. This consititutes one cycle. The number of cycles shall be three. The tests of 4.2.3 shall be conducted during each cycle while at altitude. The units shall be operationally tested within 1 hour after the end of the last test cycle. Post tests shall be performed in accordance with 4.2.3.

4.5.5.7 Acoustical noise (operating). The TMX section shall be subjected to acoustical noise in accordance with MIL-STD-810, Method 515, Procedure I. The spectrum shall be in accordance with Figure 515.2-2 of MIL-STD-810. The overall level shall be 160 dB referenced to 0.0002 dynes

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per square centimeter. Operation shall consist of the application of power. The test time shall be 30 minutes. Post tests shall be performed in accordance with 4.2.3.

4.5.5.8 Transportation vibration (nonoperating). The TMX section shall be subjected to sinusoidal vibration in accordance with MIL-STD-810, Method 514, Procedure X, except the inputs shall be modified as follows:

Frequency (Hz)	Input
5-7	0.5 inch
8-50	1.3 g
51-100	0.01 inch
101-500	5 g

The test time shall be 84 minutes per axis (252 minutes total). Post tests shall be performed in accordance with 4.2.3.

4.5.5.9 Salt fog (nonoperating). The TMX section shall be subjected to salt fog in accordance with MIL-STD-810, Method 509, Procedure I. Post tests shall be performed in accordance with 4.2.3

4.5.5.10 Aerodynamic heating (nonoperating). The TMX section shall be subjected to one cycle of aerodynamic heating in accordance with Figure 11. Temperatures specified shall be TMX skin temperatures. Post tests shall be performed in accordance with 4.2.3.

4.5.5.11 Degree of enclosure (nonoperating). The TMX section shall be subjected to the splash proof test of MIL-STD-108. Post tests shall be performed in accordance with 4.2.3.

4.5.6 Visual examinations. The TMX subassemblies shall be visually examined to verify conformance to the requirements of 3.3.1, 3.5, and Drawing 639AS4900 and associated documents.

4.6 Test documentation.

4.6.1 Test procedures. When specified in the contract or purchase order (see 6.2.2), test procedures shall be prepared for the contracting activity for approval prior to testing.

4.6.2 Logbook and quality assurance (QA) data sheets. When specified in the contract or purchase order (see 6.2.2), a logbook and QA data sheets shall be prepared in accordance with Figure 6. The contractor's test results shall be recorded in accordance with the example QA data sheet shown in Figure 12.

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4.6.3 Certification. When specified in the contract or purchase order (see 6.2.2), the contractor shall have available for the Government representative written certification, accompanied by objective quality evidence in accordance with MIL-STD-109, that the components and parts used in the TMX section meet the applicable requirements.

4.7 Packaging and packing. Prior to shipment, packaging and shipping containers shall be inspected to determine conformance with the applicable requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-STD-794, Level C.

5.1.1 Level C. The TMX section shall be packaged in suitable commercially-available materials in a manner that will provide adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packaging shall be in accordance with 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with MIL-STD-794, Level C.

5.2.1 Level C. The TMX section, packaged in accordance with 5.1, shall be packed in suitable commercially-available materials that will provide adequate protection against damage during direct shipment from the supply source to the first receiving activity. Containers shall be packed in accordance with 49 CFR 171-178.

5.3 Marking. In addition to any special markings required by the contract or purchase order (see 6.2.1), unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 and 49 CFR 171-178.

6. NOTES.

6.1 Intended use. The AN/DKT-58(V1) Telemetric Section Transmitting Set is an airborne guided missile transmitting telemetric section, used to sample and transmit flight performance information for a 5-inch guided missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Transmitter center frequency (see 3.2.1.5).

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c. Serial numbers (see 3.3.1.1).

d. MTBF demonstration (see 3.2.3.1).

e. Working environment (see 3.3.2.1).

f. Responsibility for inspection and inspection facilities, if other than specified in 4.1.

g. Calibration (see 4.4.1).

h. Test equipment approval (see 4.4.2).

i. Marking (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.3	Reliability Program Plan	DI-R-7079	---
4.4.1	Certification	DI-E-2121	---
4.6.1	Test Procedures	UDI-T-21347	---
4.6.2	QA Data Sheets	UDI-R-23573	---
4.6.3	Logbook	UDI-M-21271	---
4.6.3	Certification	DI-E-2121	---

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of four units (see 4.2). The contracting officer

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should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the document's first article.

6.3 Definitions. The following definitions apply to this specification. (see Figure 2).

6.3.1 Acquisition window. The acquisition window is that portion of the pulse that constantly remains between 99 and 101 percent of 100 percent amplitude.

6.3.2 Input onset. The time of the start of the input pulse.

6.3.3 Propagation delay. The time from the input onset to the output onset.

6.3.4 Output onset. The actual time of the start of the initial output rise.

6.3.5 Leading edge. The initial rise of the pulse.

6.3.6 Period. The time between the beginning and the termination of the event.

6.3.7 Risetime. Risetime is measured from the output onset to the start of the acquisition window.

6.3.8 Overshoot. That portion of the curve that first extends over the 100 percent level.

6.3.8.1 Switching transient overshoot. An overshoot that occurs between two data sampling intervals.

6.3.9 Ringtime. Ringtime is between the time 100 percent amplitude is first attained and the time at which all ringing falls within the acquisition window.

6.3.10 Tilt. The amount of rise or drop from the horizontal 100 percent level measured in percent per microsecond.

6.3.11 Trailing edge. The fall due to termination of the input event. All characteristics of the trailing edge have the same definitions as the leading edge.

Preparing activity:
NAVY-AS
(Project 1420-N263)

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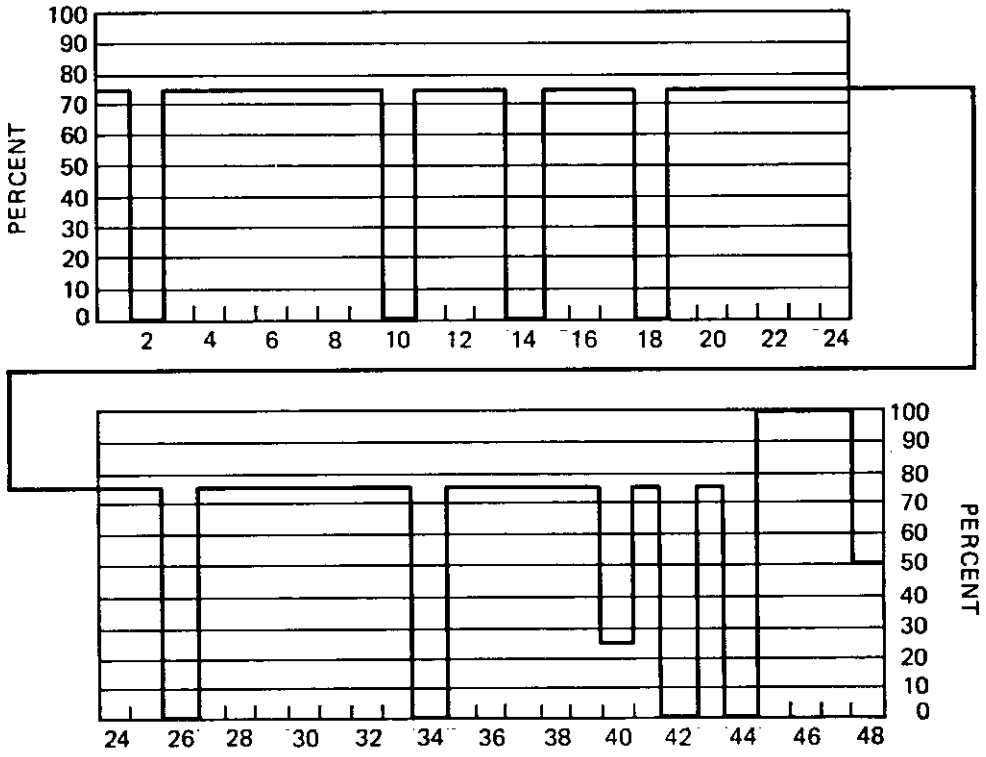


FIGURE 1. PAM output for AN/DKT-58 (V1).

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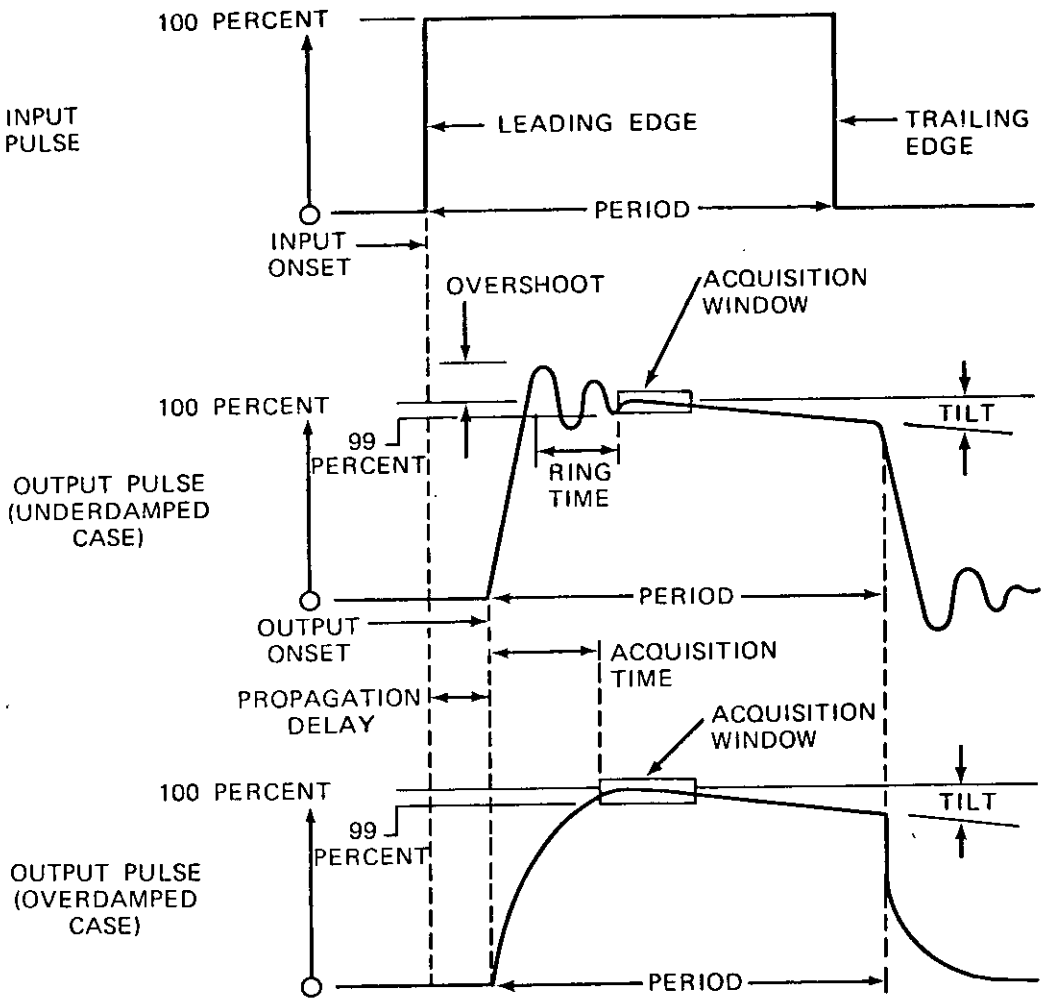


FIGURE 2. Input and output pulse characteristics.

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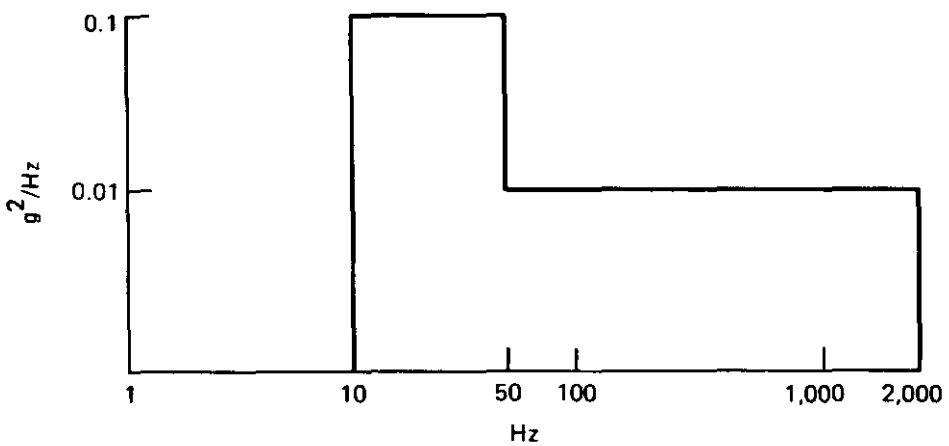
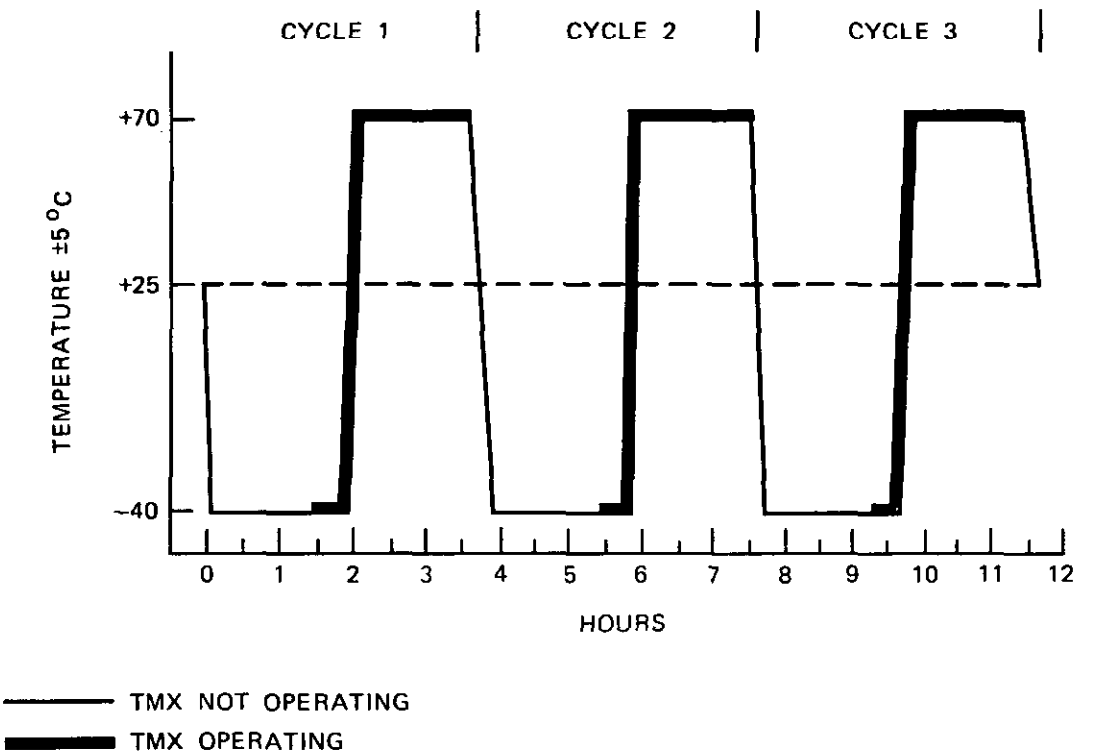


FIGURE 3. Random vibration.

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NOTES:

1. Each TMX shall be subjected to not less than three temperature cycles.
2. The temperature rate of change shall be not less than 10°C per minute.
3. Temperature soak times shall be 2 hours at –40 ±5°C and 1.5 hours at +70 ±5°C.
4. The TMX shall be operating 30 minutes prior to the start of the heating cycle and continue to operate through the 1.5 hours soak time at +70 ±5°C.
5. Any TMX that fails during burn-in temperature cycling shall be reworked and resubmitted to the temperature cycling screen at the beginning of the cycle preceding the cycle the TMX failed.

FIGURE 4. TMX burn-in/temperature cycling profile.

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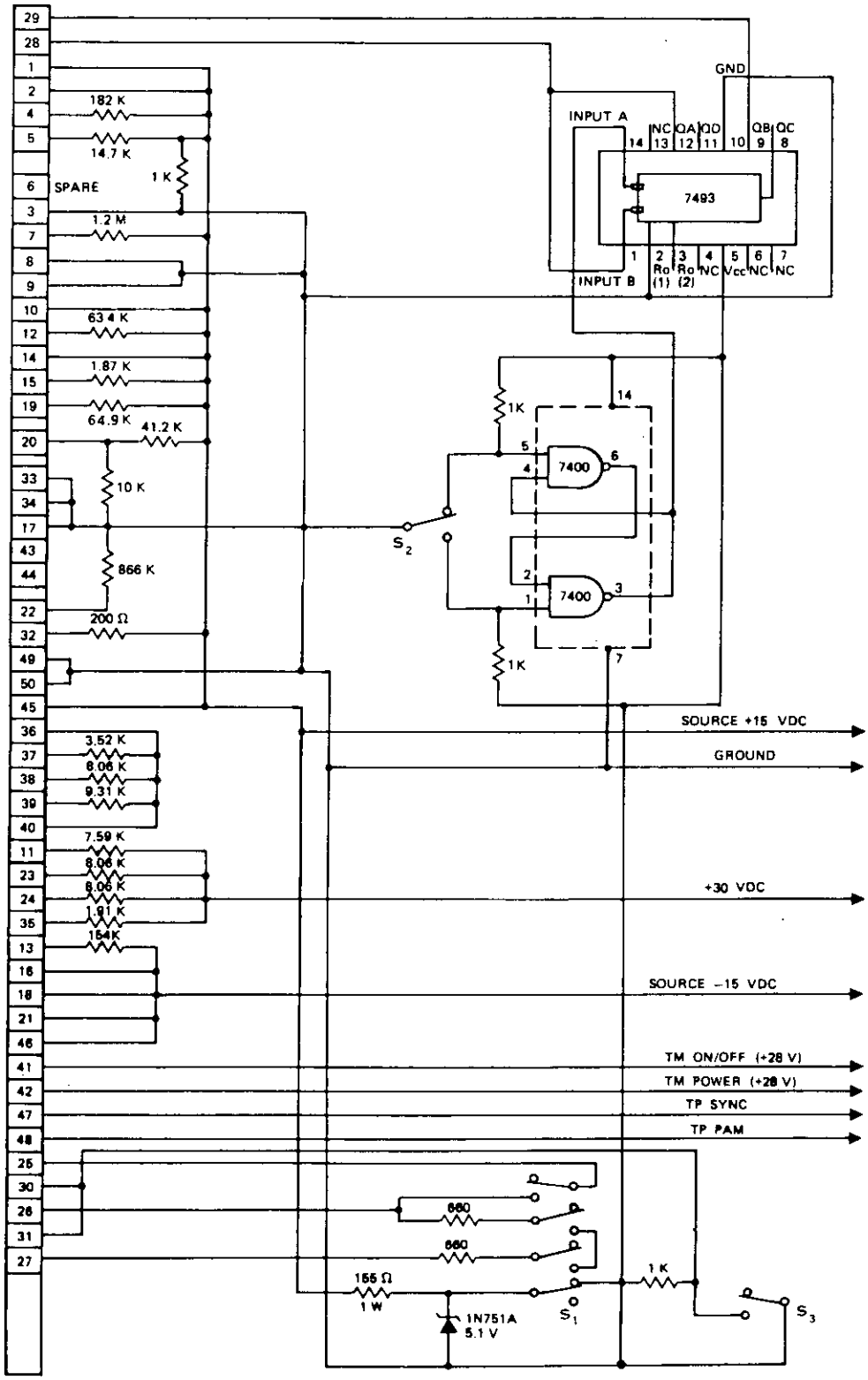


FIGURE 5. AN/DKT-58(V1) programmable connector and associate control circuitry.

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LOGBOOK

FOR

TELEMETRIC SECTION

GUIDED MISSILE

AN/DKT-58(V1)

MFD BY:_____ CONTRACT NO._____

SERIAL NO._____ DATE 1ST ATP:_____

DATE RECEIVED:_____

FREQ: _____

VISUAL AND MECHANICAL INSPECTION PERFORMED _____

CONTINUITY MEASUREMENTS PERFORMED _____

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1).

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MAJOR COMPONENTS LIST Serial No. _____

Component	NAVAIR (Latest Spec/Dwg Rev.)	Manufacturer	Manufacturer's Model No.	Serial Number

Remarks:

Verified_____

Date_____

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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AN/DKT-58 Quality Assurance Data Sheets Serial No. _____				Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units					
1.	4.5.1.2	TMX turns on/off properly	OK					
2.	4.5.1.3	Source current transmitter off Limits: 350 maximum	mA					
3.	4.5.1.4	Source current transmitter on Limits: 1.5 maximum	amperes					
4.	4.5.1.5	Transmitter power output Limits: 3.0 W minimum at ambient 2.0 W minimum all other environments	watts					
5.	4.5.1.6	Transmitter center frequency Limits: Center frequency ±0.003 percent	MHz					
6.	4.5.1.7a	Observe PAM output on scope	OK					
7.	4.5.1.7c	Channel voltage levels Channels Limits 1 1.00 to 1.50	Vdc					
		3 1.00 to 1.50	Vdc					
		4 1.00 to 1.50	Vdc					
		5 1.00 to 1.50	Vdc					
		6 1.00 to 1.50	Vdc					
		7 1.00 to 1.50	Vdc					
		8 1.00 to 1.50	Vdc					
		9 1.00 to 1.50	Vdc					

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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AN/DKT-58 Quality Assurance Data Sheets Serial No. _____				Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units					
7.	4.5.1.7c (continued)	Channels Limits						
		11 1.00 to 1.50	Vdc					
		12 1.00 to 1.50	Vdc					
		13 1.00 to 1.50	Vdc					
		14 -2.40 to -2.55	Vdc					
		15 1.00 to 1.50	Vdc					
		16 1.00 to 1.50	Vdc					
		17 1.00 to 1.50	Vdc					
		19 1.00 to 1.50	Vdc					
		20 1.00 to 1.50	Vdc					
		21 1.00 to 1.50	Vdc					
		22 1.00 to 1.50	Vdc					
		23 1.00 to 1.50	Vdc					
		25 1.00 to 1.50	Vdc					
		27 1.00 to 1.50	Vdc					
		28 1.00 to 1.50	Vdc					
		29 1.00 to 1.50	Vdc					
		30 1.00 to 1.50	Vdc					
		31 1.00 to 1.50	Vdc					
		32 1.00 to 1.50	Vdc					
		33 1.00 to 1.50	Vdc					

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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AN/DKT-58 Quality Assurance Data Sheets Serial No. _____				Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units					
7.	4.5.1.7c (continued)	Channels Limits						
		35 1.00 to 1.50	Vdc					
		36 1.00 to 1.50	Vdc					
		37 1.00 to 1.50	Vdc					
		38 1.00 to 1.50	Vdc					
		39 1.00 to 1.50	Vdc					
		40 -1.00 to -1.50	Vdc					
		41 1.00 to 1.50	Vdc					
		43 1.00 to 1.50	Vdc					
		44 Zero ref -2.48 to -2.53	Vdc					
		45 Full scale +2.48 to +2.53	Vdc					
		46 Full scale 2.48 to 2.53	Vdc					
		47 Full scale 2.48 to 2.53	Vdc					
		48 50 percent ref 0.05 to -0.05	Vdc					
8.	4.5.1.7d,e	Channels 2, 10, 18, 26, 34, and 42 verified for conformance						
		S_1 normally open PAM output limits:						
		-2.48 to -2.53 Vdc ≃ 0 percent	OK					
		S_1 normally closed, S_3 normally open, toggle S_2						

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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AN/DKT-58 Quality Assurance Data Sheets		Serial No. _____		Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units					
8.	4.5.1.7d,e (continued)	PAM output limits State 1: 1.43 to 1.59 Vdc ≃ 82 percent State 2: 0.43 to 0.46 Vdc ≃ 60 percent State 3: -0.43 to -0.47 Vdc ≃ 41 percent State 4: -1.39 to -1.53 Vdc ≃ 20 percent	 OK OK OK OK					
		S_1 normally closed, S_3 normally closed, toggle S_2 PAM output limits State 1: 1.43 to 1.59 Vdc ≃ 82 percent State 2: -0.43 to -0.47 Vdc ≃ 41 percent	 OK OK					
9.	4.5.1.8c	Transmitter upper band edge frequency Limits: 240 to 270 kHz	kHz					
10.	4.5.1.8d	Transmitter lower band edge frequency Limits: 240 to 270 kHz	kHz					
11.	4.5.1.8e	Peak-to-peak deviation Limits: 480 to 540 kHz	kHz					
12.	4.5.1.9	Risetime: Leading edge of any channel shall be not greater than 2.0 µsec Switching transient overshoot: Shall not exceed 5 percent of the scale and the duration shall not exceed 1 µsec	µsec percent					

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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AN/DKT-58 Quality Assurance Data Sheets Serial No. _____				Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units					
13.	4.5.1.10 a.	Squib test Observe channel 40 TM squib indicator Limits: -1 to -1.5 Vdc	Vdc					
	c.	Thermal battery tests Limits: 0.55 to 0.70 ohms	ohms					
	d.	Launch latch test Measure diode resistance between P1-44 and P1-49. Limits: 0 to 300	ohms					
	e.	Reverse squib meter leads Limits: ∞	OK					
14.	4.5.1.11	Airborne power supplies +15 volts Limits: +14.93 to +15.07 Vdc -15 volts Limits: -14.93 to -15.07 Vdc Date _____ Operator _____ Date _____ QA (Contractor) _____ Date _____ DCAS _____	Vdc Vdc					

FIGURE 6. Logbook for telemetric section guided missile AN/DKT-58(V1) - continued.

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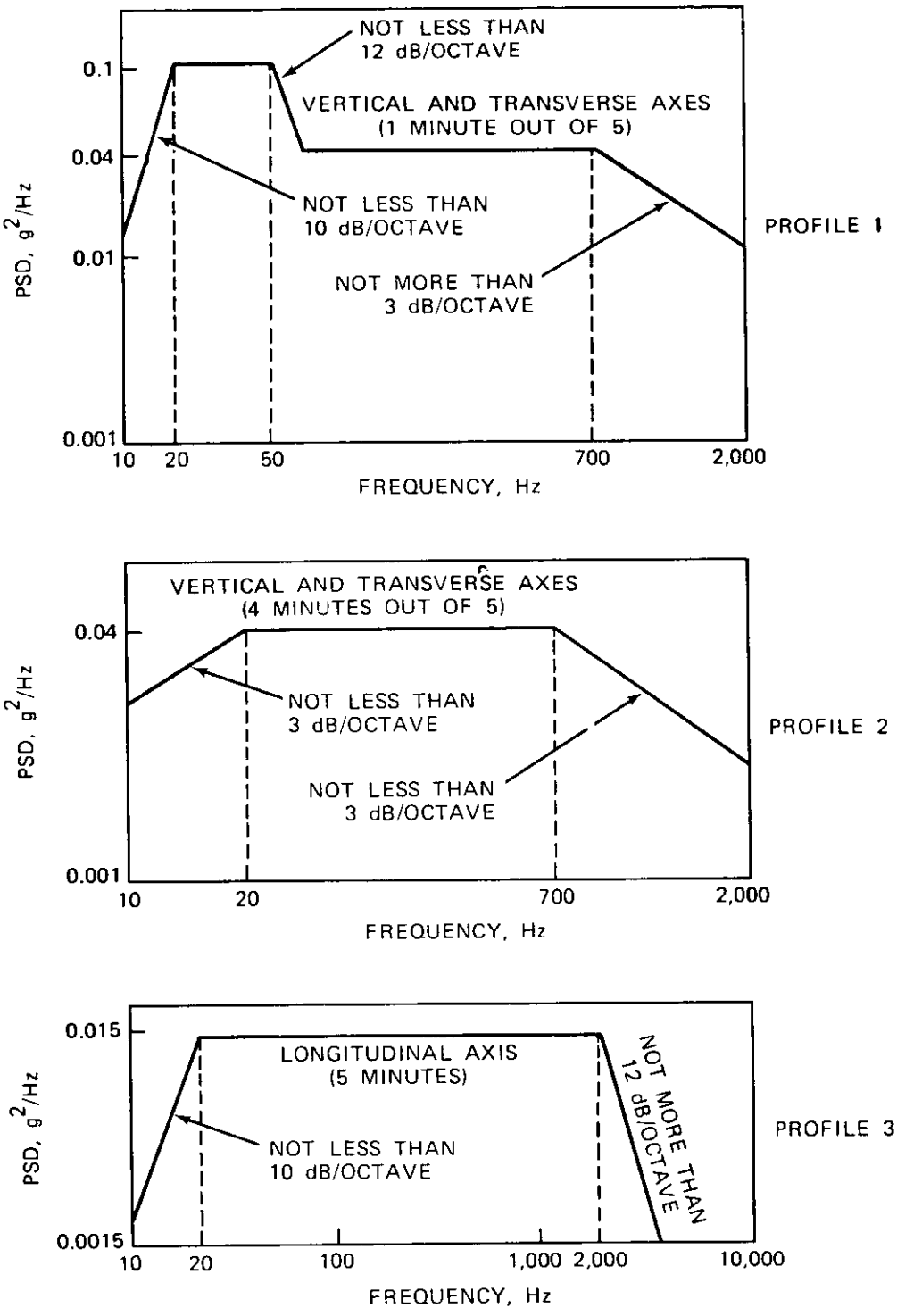
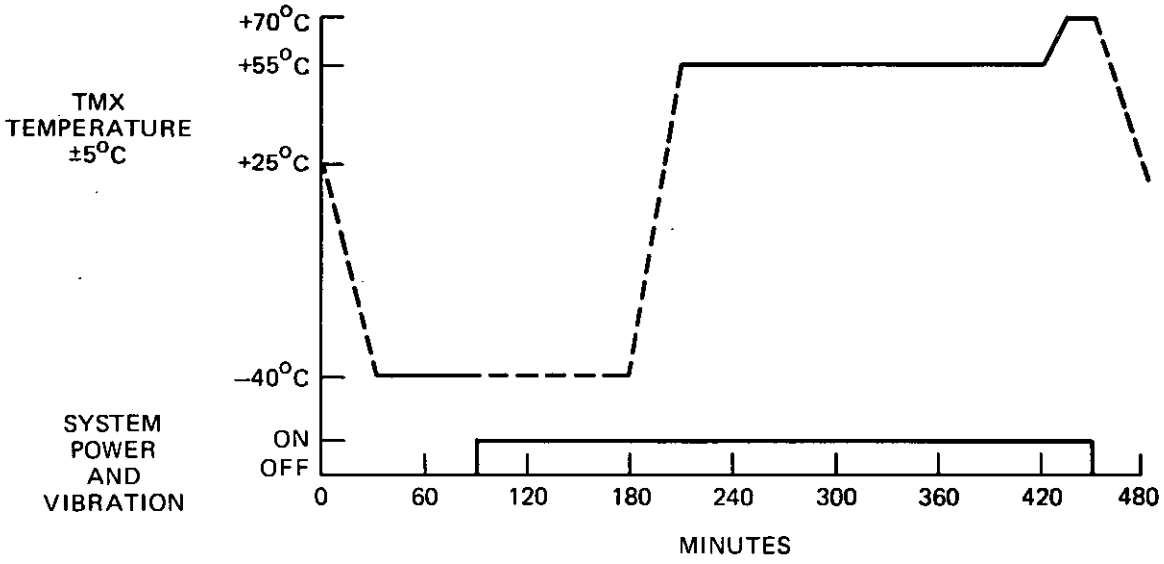


FIGURE 7. TMX random vibration screen at ambient.

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MTBF TEST CYCLE

STEP	SEQUENTIAL TEST TIME PERIOD (MINUTES) T	TIME DURATION (MINUTES)	TEMPERATURE ACTIVITY	TEMPERATURE REQUIREMENT ±5°C	SYSTEM POWER		VIBRATION
					ON TIME	OFF TIME	
1	0 TO 30	30	TRANSITION	+25° TO –40°		30	OFF
2	30 TO 90	60	DWELL	–40°		60	OFF
3	90 TO 180	90	MAINTAIN CHAMBER	< –40°	90		ON
4	180 TO 210	30	TRANSITION	TO +55°	30		ON
5	210 TO 420	210	DWELL	+55°	210		ON
6	420 TO 435	15	TRANSITION	TO +70°	15		ON
7	435 TO 450	15	DWELL	+70°	15		ON
8	450 TO 480	30	TRANSITION	TO +25°		30	OFF
TOTAL		480			360	120	

FIGURE 8. MTBF testing for Fleet TMX.

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MTBF VIBRATION PROFILES

PROFILE	INPUT PSD LEVEL, g^2/Hz	FREQUENCY, Hz	DURATION, MINUTES
A	0.005	20 TO 2,000	CONTINUOUS
B	0.015	20 TO 2,000	3
C	0.05	20 TO 100	1

MTBF VIBRATION APPLICATION

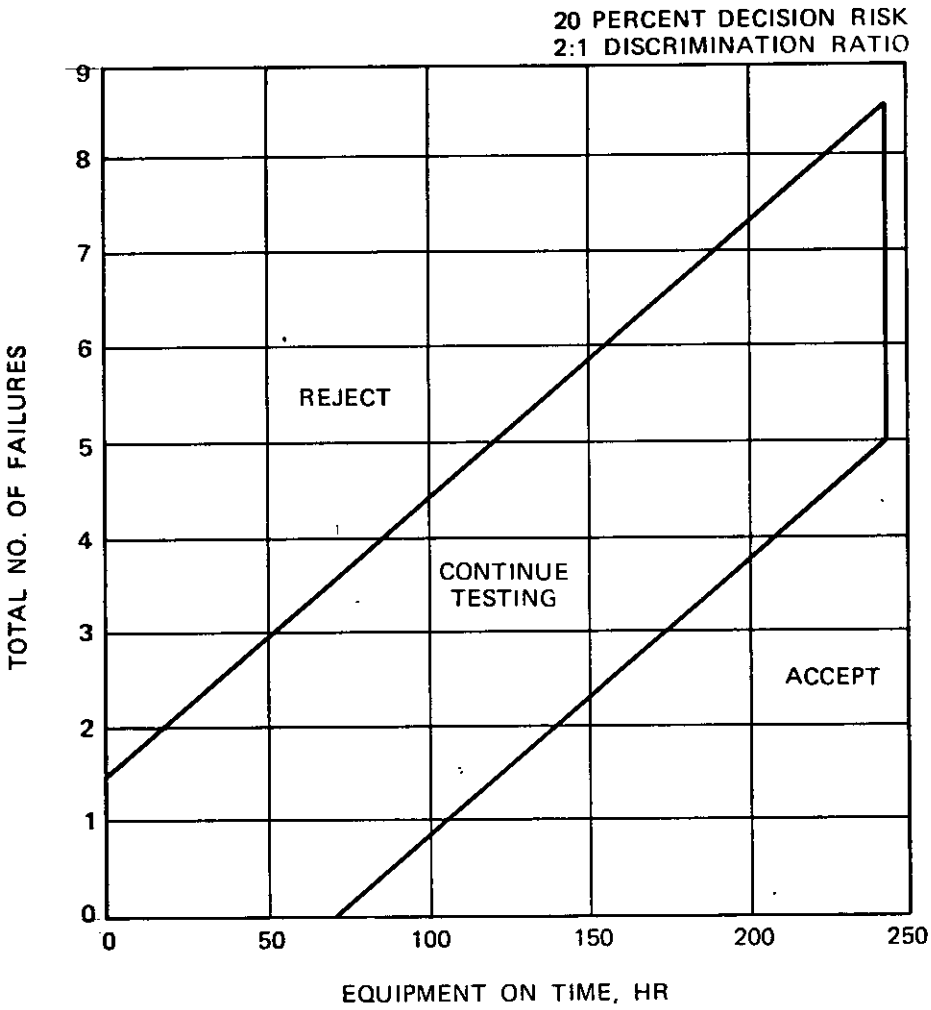
T ± 5 MINUTES	143	170	197	224	251	278	305	332	358	386	413	440
PROFILE	C	B	C	C	B	B	B	C	B	C	B	C

NOTES:

1. System power input voltage shall be in accordance with the power input voltage requirement as specified herein.
2. Chamber temperature during step 3 shall be maintained below $-40^{\circ}C$.
3. A performance test shall be made every 6 ±3 test cycles and at the completion of required test hours.
4. Vibration profile A shall be applied continuously during the test cycle in which the TMX is energized. Vibration profiles B and C shall be applied in accordance with the MTBF vibration application. Input vibration level shall be monitored on the vibration fixture. Transition time from one vibration profile to another vibration profile shall be with minimal transitory time. Vibration shall be through the longitudinal axis of the TMX.

FIGURE 8. MTBF testing for Fleet TMX - continued.

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NUMBER OF FAILURES	REJECT TEST TIME	ACCEPT TEST TIME
0	N/A	70.0
1	N/A	104.5
2	17.5	139.5
3	52.0	174.0
4	86.5	208.5
5	121.5	243.5
6	156.0	243.5
7	190.5	243.5
8	243.5	N/A

FIGURE 9. MTBF accept/reject total test time for MTBF θ_0 of 50 hours.

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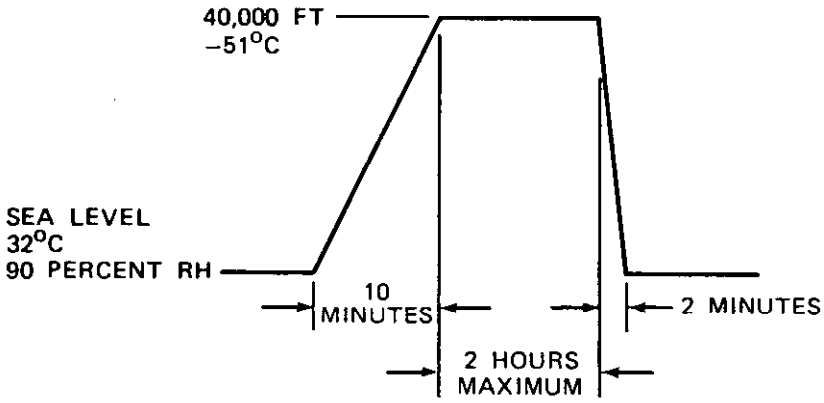


FIGURE 10. Temperature/humidity/altitude profile.

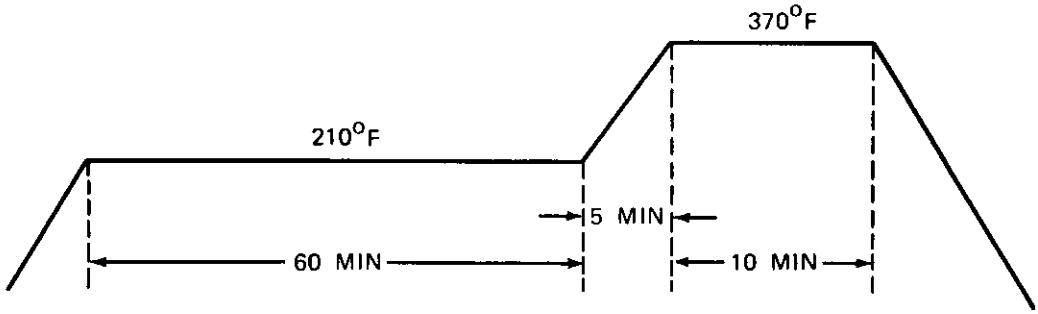


FIGURE 11. Aerodynamic heating profile.

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AN/DKT-58 Quality Assurance Data Sheets Serial No. _____				Test runs				
Item no.	MIL-T-85574 paragraph	Test and conditions	Units	Ambient 1/5/83	High temp 1/11	Cold temp 1/12	Vib 1/14	
1.		TMX turns on/off properly	OK					
2.		Source current transmitter off Limits: 350 maximum	mA					
3.		Source current transmitter on Limits: 1.5 maximum	amperes					

FIGURE 12. Example quality assurance data sheet.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. Recommended Wording:

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